# Alexander Escamilla

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## RESEARCH EXPERIENCE

**SURP and MESA Researcher – San Diego State University** **Sep 2024~Sep 2025**

- Designed and led an independent senior-thesis project expanding prior CASSI work into a comprehensive study on feedback-regulated star formation.
- Developed a synthetic-observation pipeline integrating radiative-transfer modeling, FIR dust-emission mapping (150–500 micrometers), and a young stellar object pipeline for accretion and selection.
- Applied statistical analysis using **Python** to quantify how feedback physics, spatial resolution, and observational assumptions bias inferred star-formation efficiencies across evolving giant molecular clouds.

**Summer Research Intern - Carnegie Observatories** **Jun 2024~Aug 2024**

- Collaborated with Dr. Mike Grudić ( FlatIron Institute) and Dr. Rosen to compare theoretical and observational estimates of star formation efficiency in giant molecular clouds.
- Modeled far-infrared dust emission via radiative transfer and created mock STARFORGE observations to bridge discrepancies between theory and observation.

**Research Assistant - San Diego State University** **Jan 2024~May 2024**

- I joined Dr. Anna Rosen's computational astrophysics group to study how stellar winds and feedback regulate star formation in the interstellar medium using STARFORGE simulations.
- Learned simulation analysis, developed independence in research through self-study and mentorship

**Undergraduate Researcher – STARS Program | San Diego State University** **Jun 2022 ~ Aug 2022**

● Conducted introductory astrophysics research analyzing dark-matter distributions in galaxies under faculty mentorship, gaining foundational coding and data-analysis skills.

## PUBLICATIONS & CONFERENCES

**Peer-Reviewed Paper**

- A. Escamilla, M. Y. Grudić, & A. L. Rosen (2025). *"Bridging Theory and Observation: Synthetic FIR Insights into Star Formation Efficiency." The Astrophysical Journal, Volume 1004, Number 1.* *https://iopscience.iop.org/article/10.3847/1538-4357/ae6cdc*

**Conference Presentations & Talks**

- **GalFRESCA 2025 Flash Talk (Sep 2025):** Bridging Theory and Observation: Synthetic FIR Insights into Star Formation Efficiency.
- **American Astronomical Society (AAS) 245th Winter Meeting (Jan 2025)**: **IPoster** — Measuring Star Formation Efficiency in STARFORGE
- **Ensenada San Diego Astrophysical Society Meeting (Sep 2024)**: **Talk** — Measuring Star Formation Efficiency in STARFORGE
- **Carnegie Cassi Research Symposium (Aug 2024): Talk -** Measuring Star Formation Efficiency in STARFORGE

## EDUCATION

**San Diego State University (SDSU)** **May 2025**

Major: Astronomy

- **Honors & Awards**: Cum Laude, Dean's List (Multiple Semesters), LSAMP and MESA Scholar, SURP Research Grant Recipient - Funded senior-thesis research
- **Major GPA:** 3.67
- **Relevant Coursework:** Stellar Astrophysics, Computational Modeling, Thermodynamics, Quantum Mechanics, Astronomical Techniques

**San Diego Community College District (SDCCD)** **June 2023**

Program: IGETC Transfer Certificate in Physics

## TECHNICAL SKILLS

**Python (yt, Astropy, NumPy, Matplotlib, h5py, pandas) | Linux | Git | LaTeX | Scientific Writing**